Topic: Phase portraits for distinct real Eigenvalues

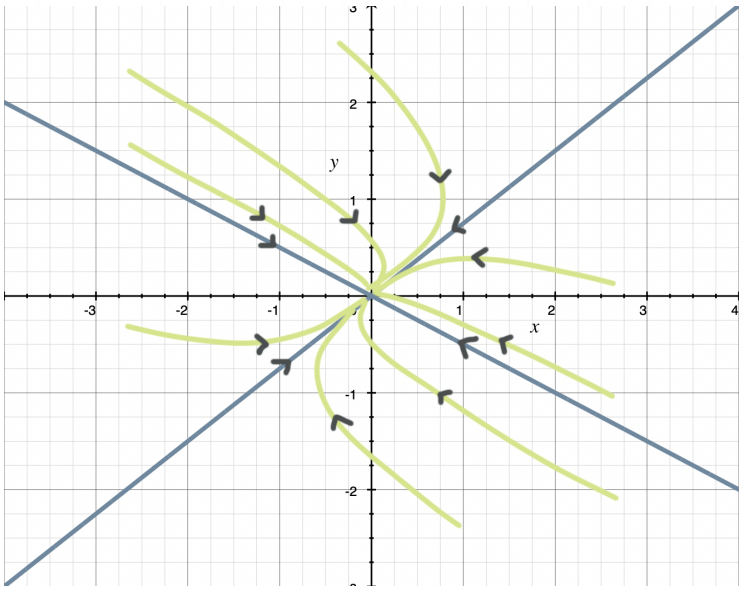
Question: Sketch the phase portrait of the system.

$$x_1' = -\frac{5}{2}x_1 + 2x_2$$

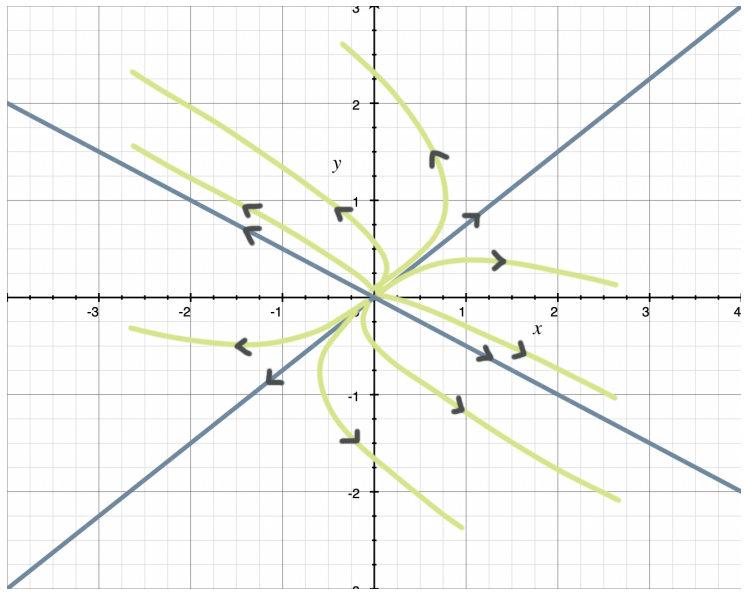
$$x_2' = \frac{3}{4}x_1 - 2x_2$$

Answer choices:

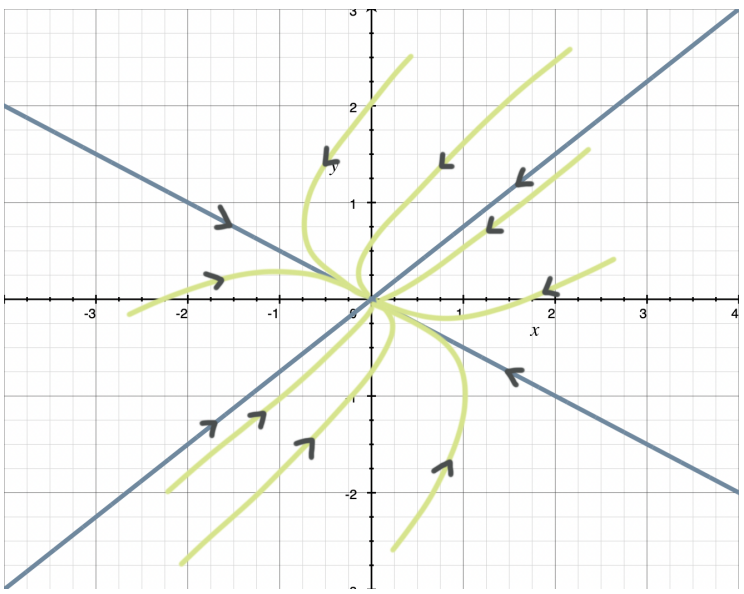
A



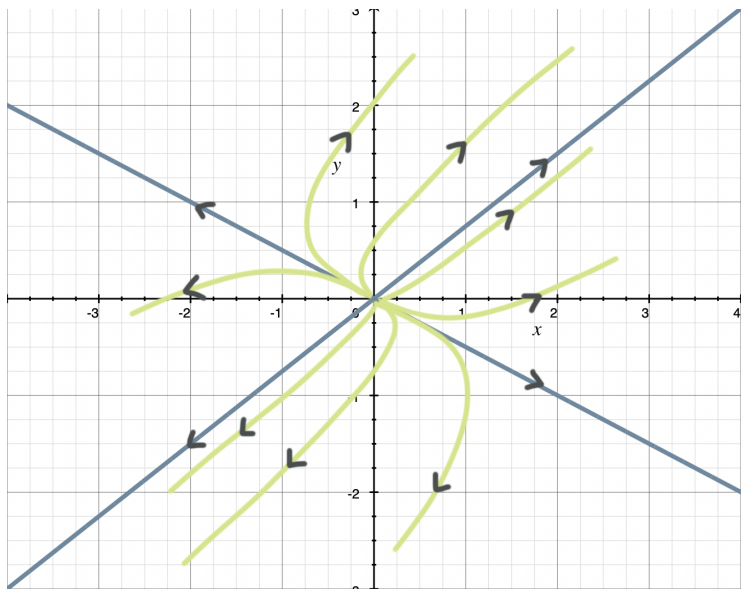
B



C



D



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Solution: A

The coefficient matrix and $A - \lambda I$ are,

$$A = \begin{bmatrix} -\frac{5}{2} & 2 \\ \frac{3}{4} & -2 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} -\frac{5}{2} - \lambda & 2 \\ \frac{3}{4} & -2 - \lambda \end{bmatrix}$$

and the characteristic equation

$$\left(-\frac{5}{2} - \lambda\right)(-2 - \lambda) - (2)\left(\frac{3}{4}\right) = 0$$

$$\lambda^2 + \frac{9}{2}\lambda + \frac{7}{2} = 0$$

$$\left(\lambda + \frac{9}{4}\right)^2 = \frac{25}{16}$$

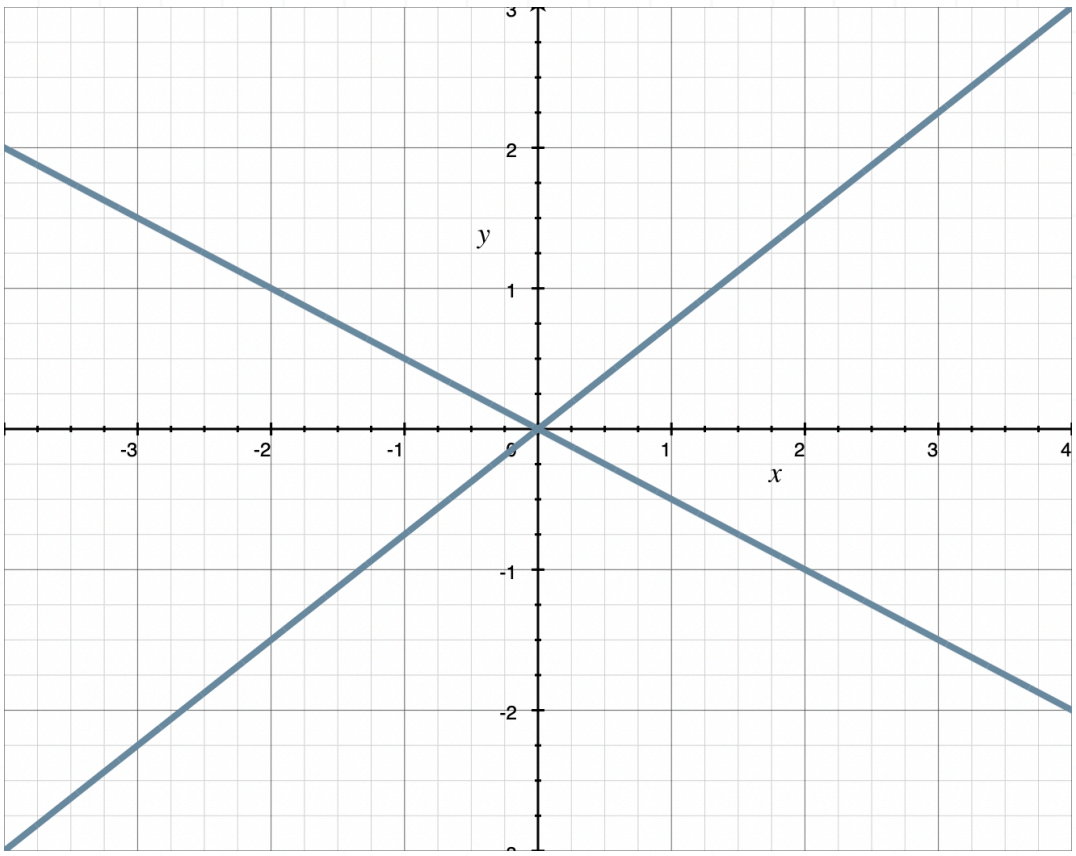
$$\lambda = -1, \ -\frac{7}{2}$$

gives the Eigenvalues $\lambda_1 = -1$ and $\lambda_2 = -7/2$, and their associated Eigenvectors

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ \frac{3}{4} \end{bmatrix} \qquad \overrightarrow{k_2} = \begin{bmatrix} 1 \\ -\frac{1}{2} \end{bmatrix}$$

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The Eigenvector $\overrightarrow{k_1} = (1,3/4)$ lies along the line $y = (3/4)x$, and the Eigenvector $\overrightarrow{k_2} = (1, -1/2)$ lies along the line $y = (-1/2)x$, so we'll sketch these lines.



The Eigenvalue associated with $\overrightarrow{k_1} = (1,3/4)$ is $\lambda = -1$, which means the direction along that trajectory is toward the origin. The Eigenvalue associated with $\overrightarrow{k_2} = (1, -1/2)$ is $\lambda = -7/2$, which means the direction along that trajectory is also toward the origin.

Because both Eigenvalues are negative, we're dealing with an asymptotically stable attractor node that attracts all trajectories.

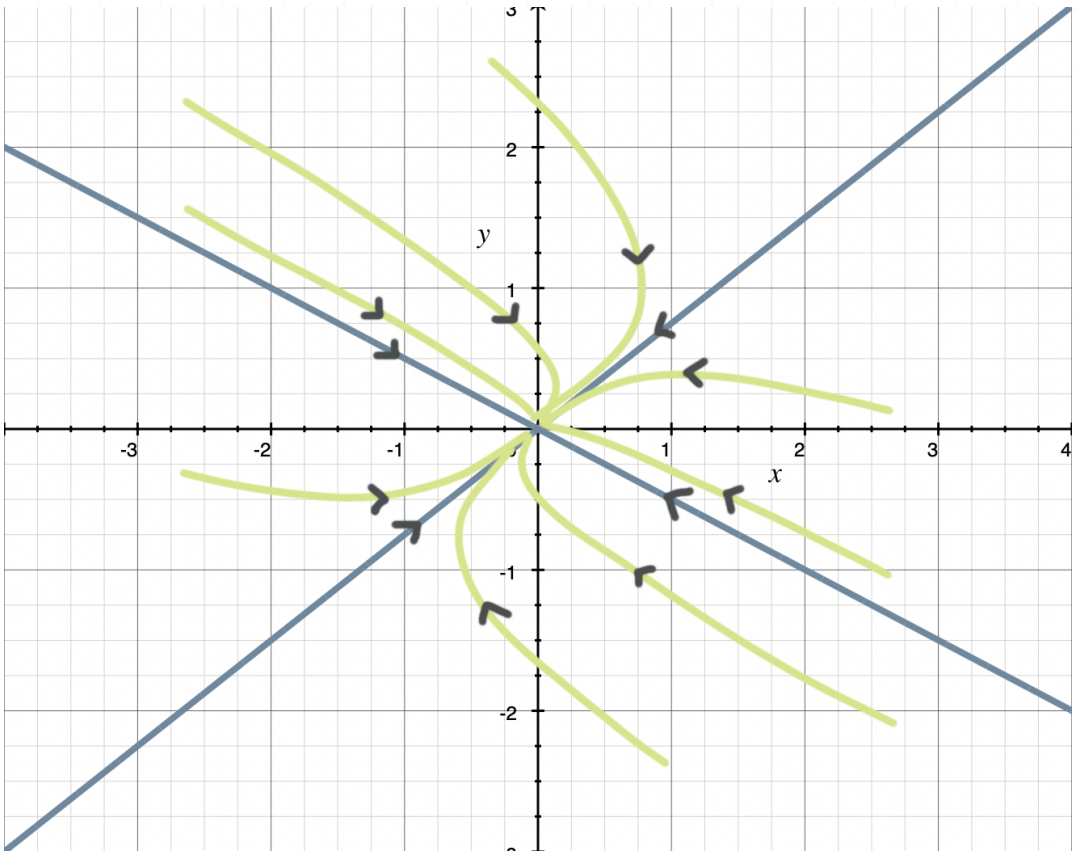
To apply the $t \to \pm\infty$ test, we'll use the general solution to the system,

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ \frac{3}{4} \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ -\frac{1}{2} \end{bmatrix} e^{-\frac{7}{2}t}$$

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Because $e^{-\frac{7}{2}t}$ dominates e^{-t} as $t \to -\infty$, the $\overrightarrow{k_1} = (1,3/4)$ vector drops away first, meaning that our trajectories are going to "start" parallel to $\overrightarrow{k_2} = (1,-1/2)$. On the other end, $e^{-\frac{7}{2}t}$ goes to 0 faster than e^{-t} as $t \to \infty$, so the $\overrightarrow{k_2} = (1,-1/2)$ vector will drop away first, meaning that our trajectories are going to "end" parallel to $\overrightarrow{k_1} = (1,3/4)$.

Therefore, starting the trajectories parallel to $\overrightarrow{k_2} = (1,-1/2)$ and ending them parallel to $\overrightarrow{k_1} = (1,3/4)$ means that the phase portrait must look something like



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Topic: Phase portraits for distinct real Eigenvalues

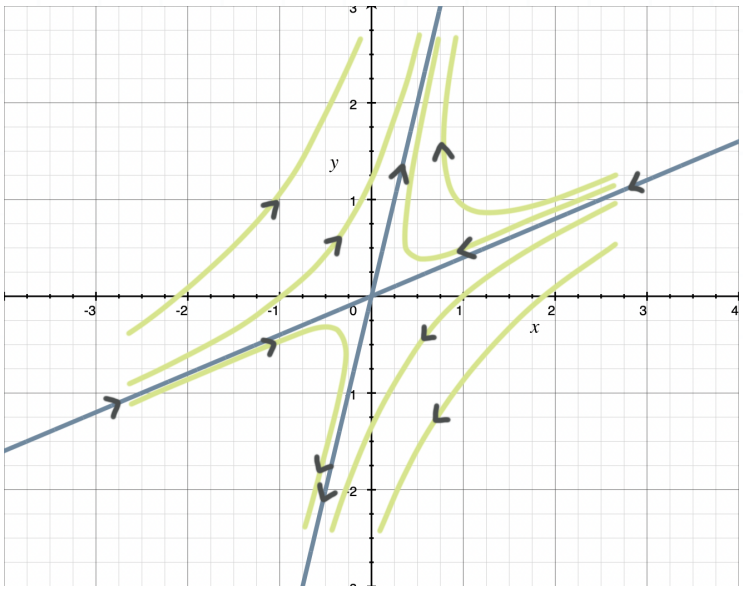
Question: Sketch the phase portrait of the system.

$$x_1' = 10x_1 - 5x_2$$

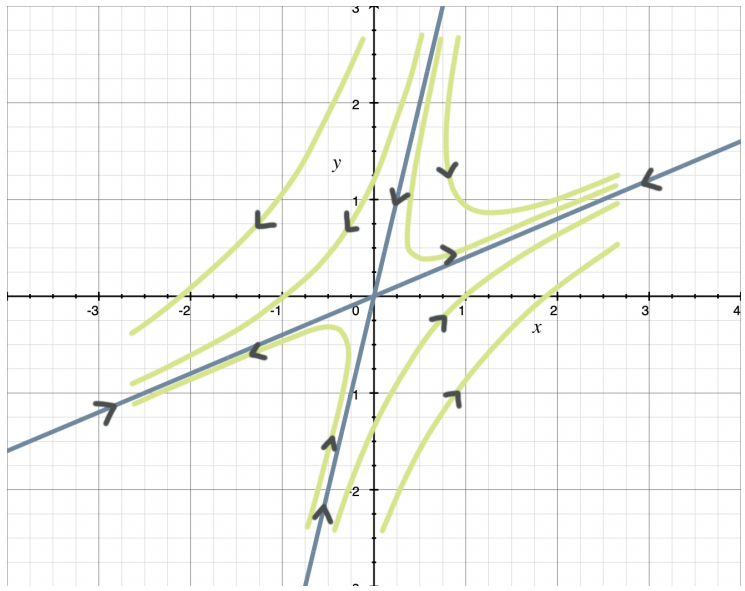
$$x_2' = 8x_1 - 12x_2$$

Answer choices:

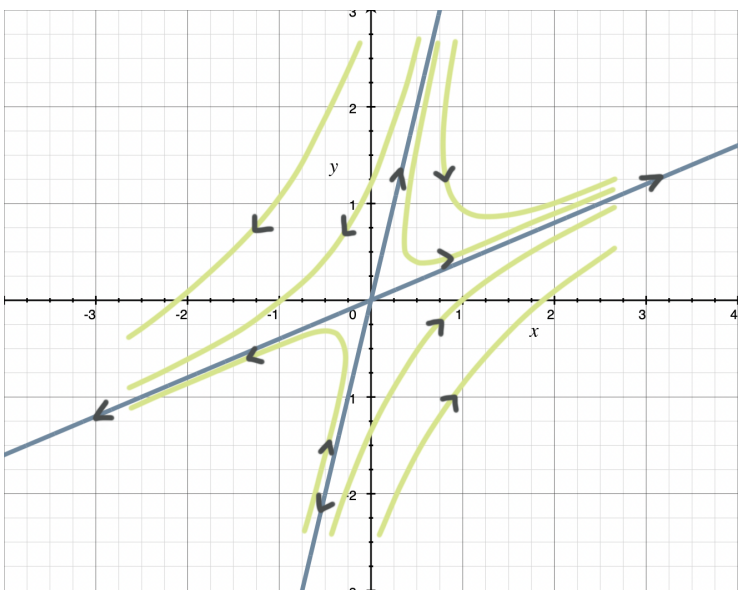
A



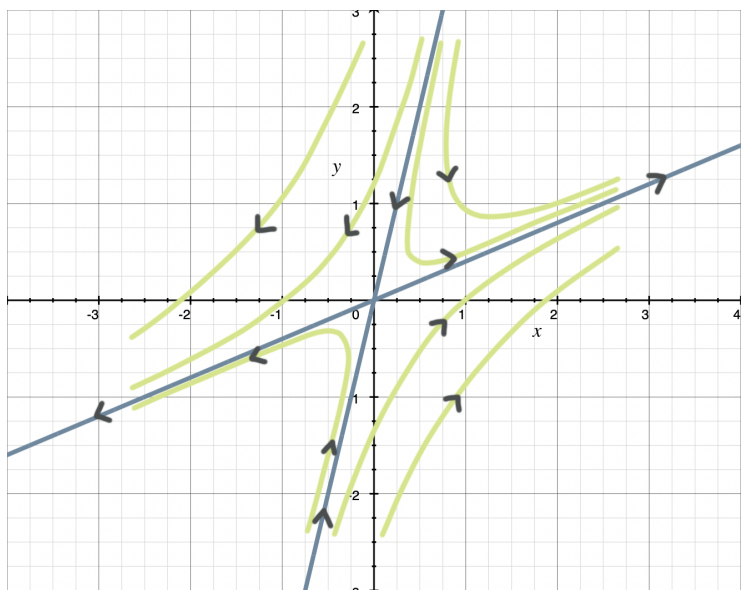
B



C



D



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Solution: D

The coefficient matrix and $A - \lambda I$ are,

$$A = \begin{bmatrix} 10 & -5 \\ 8 & -12 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 10 - \lambda & -5 \\ 8 & -12 - \lambda \end{bmatrix}$$

and the characteristic equation

$$(10 - \lambda)(-12 - \lambda) - (-5)(8) = 0$$

$$\lambda^2 + 2\lambda - 80 = 0$$

$$(\lambda + 10)(\lambda - 8) = 0$$

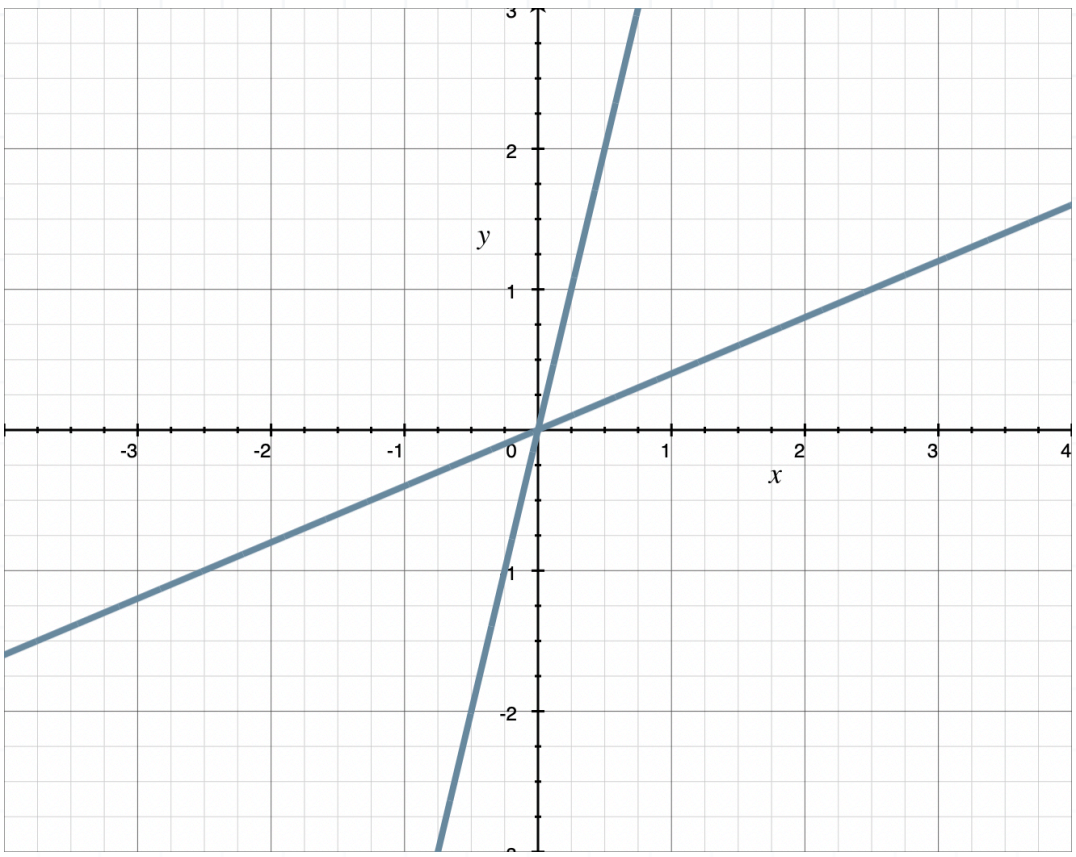
$$\lambda = -10,\ 8$$

gives the Eigenvalues $\lambda_1 = -10$ and $\lambda_2 = 8$, and their associated Eigenvectors

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 4 \end{bmatrix} \qquad \overrightarrow{k_2} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

The Eigenvector $\overrightarrow{k_1} = (1,4)$ lies along the line $y = 4x$, and the Eigenvector $\overrightarrow{k_2} = (5,2)$ lies along the line $y = (2/5)x$, so we'll sketch these lines.

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The Eigenvalue associated with $\overrightarrow{k_1} = (1,4)$ is $\lambda = -10$, which means the direction along that trajectory is toward the origin. The Eigenvalue associated with $\overrightarrow{k_2} = (5,2)$ is $\lambda = 8$, which means the direction along that trajectory is away from the origin.

Because the Eigenvalues have opposite signs, we're dealing with an unstable saddle point.

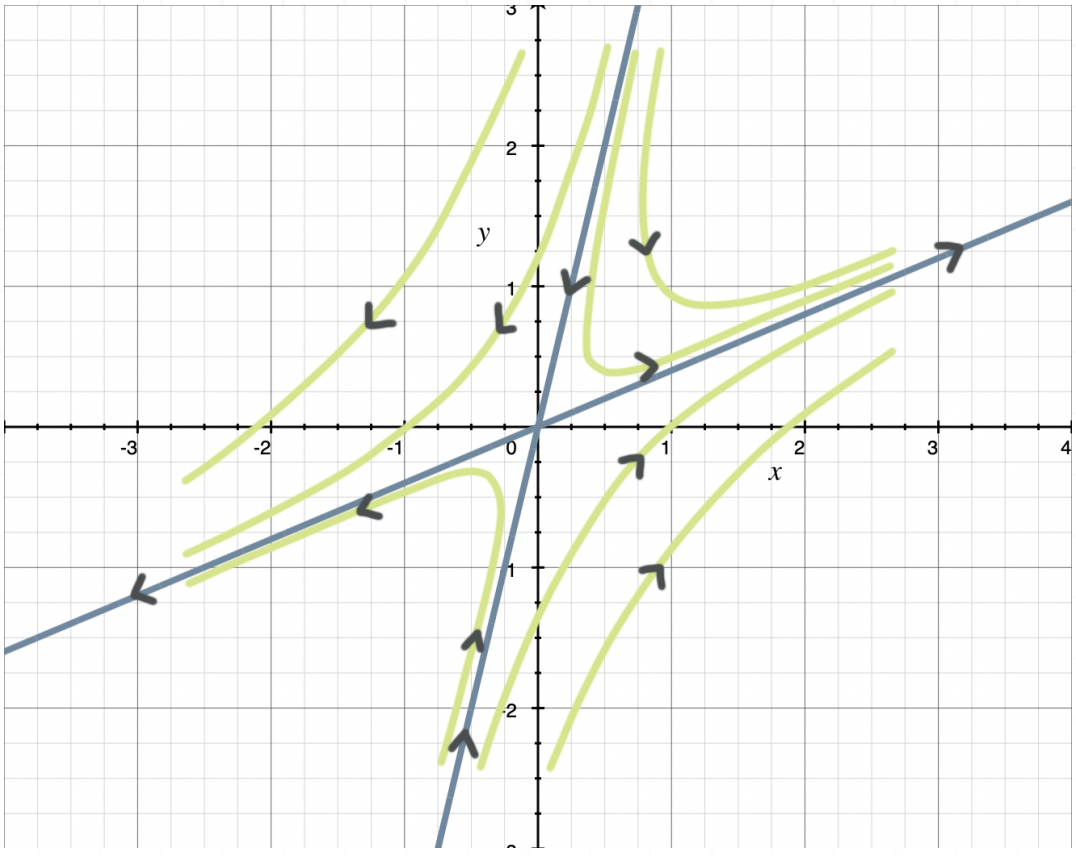
To apply the $t \to \pm\infty$ test, we'll use the general solution to the system,

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 4 \end{bmatrix} e^{-10t} + c_2 \begin{bmatrix} 5 \\ 2 \end{bmatrix} e^{8t}$$

Because e^{-10t} dominates e^{8t} as $t \to -\infty$, the $\overrightarrow{k_2} = (5,2)$ vector drops away first, meaning that our trajectories are going to "start" parallel to $\overrightarrow{k_1} = (1,4)$. On the other end, e^{8t} dominates e^{-10t} as $t \to \infty$, so the $\overrightarrow{k_1} = (1,4)$ vector will drop away first, meaning that our trajectories are going to "end" parallel to $\overrightarrow{k_2} = (5,2)$.

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Therefore, starting the trajectories parallel to $\overrightarrow{k_1} = (1,4)$ and ending them parallel to $\overrightarrow{k_2} = (5,2)$ means that the phase portrait must look something like



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Topic: Phase portraits for distinct real Eigenvalues

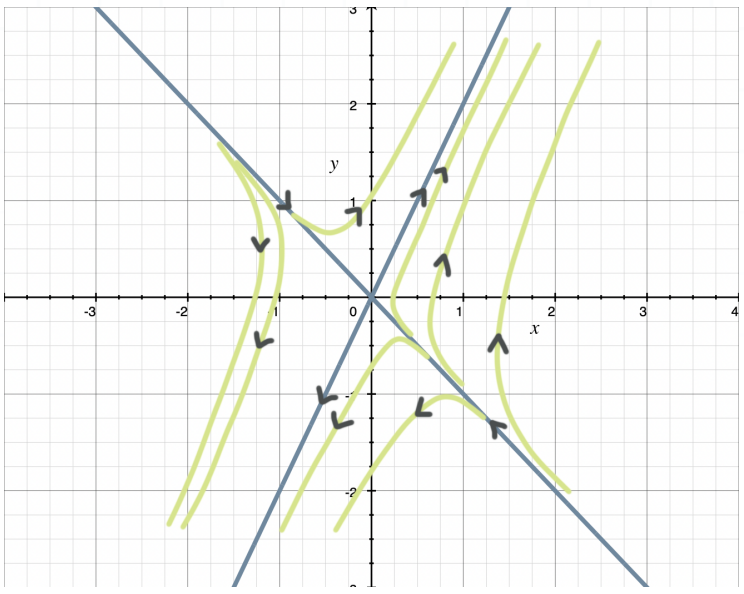
Question: Sketch the phase portrait of the system.

$$x_1' = x_1 + 2x_2$$

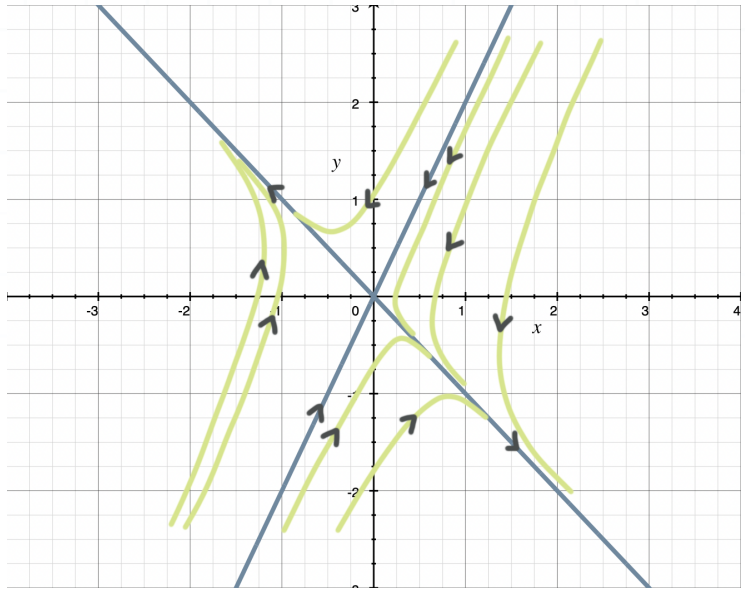
$$x_2' = 4x_1 + 3x_2$$

Answer choices:

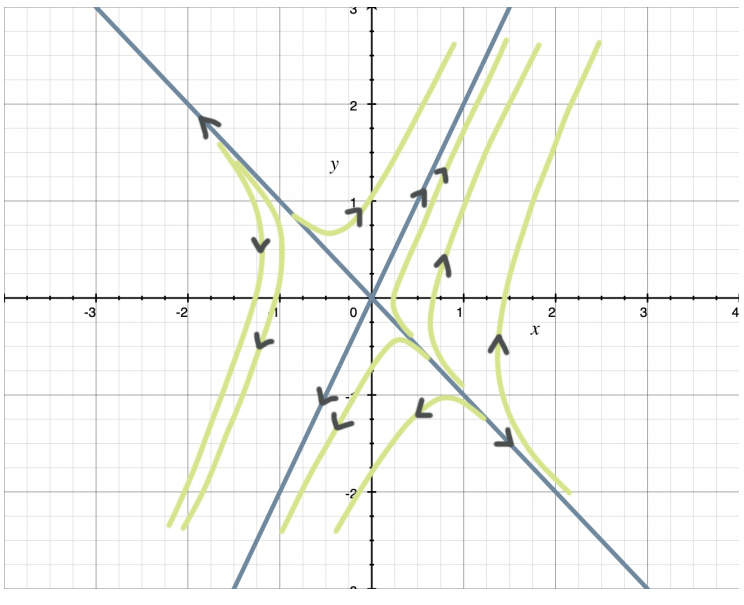
A



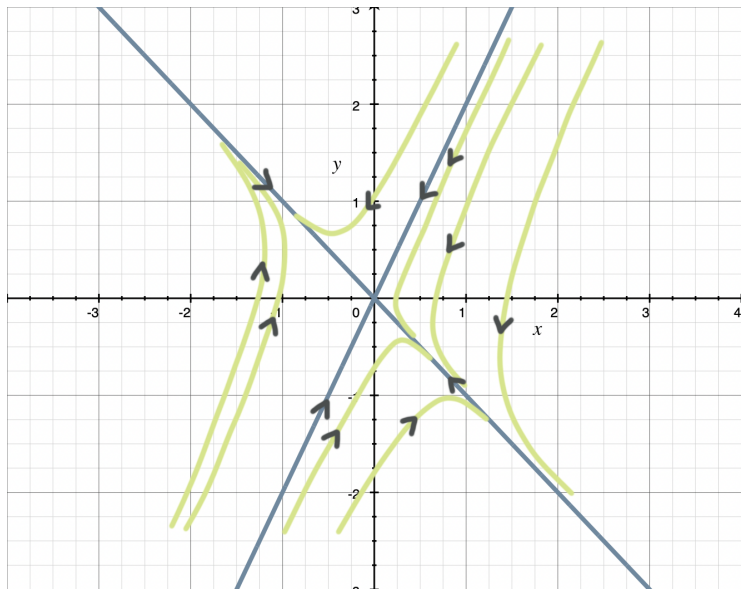
B



C



D



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Solution: A

The coefficient matrix and $A - \lambda I$ are,

$$A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 - \lambda & 2 \\ 4 & 3 - \lambda \end{bmatrix}$$

and the characteristic equation

$$(1 - \lambda)(3 - \lambda) - (2)(4) = 0$$

$$\lambda^2 - 4\lambda - 5 = 0$$

$$(\lambda + 1)(\lambda - 5) = 0$$

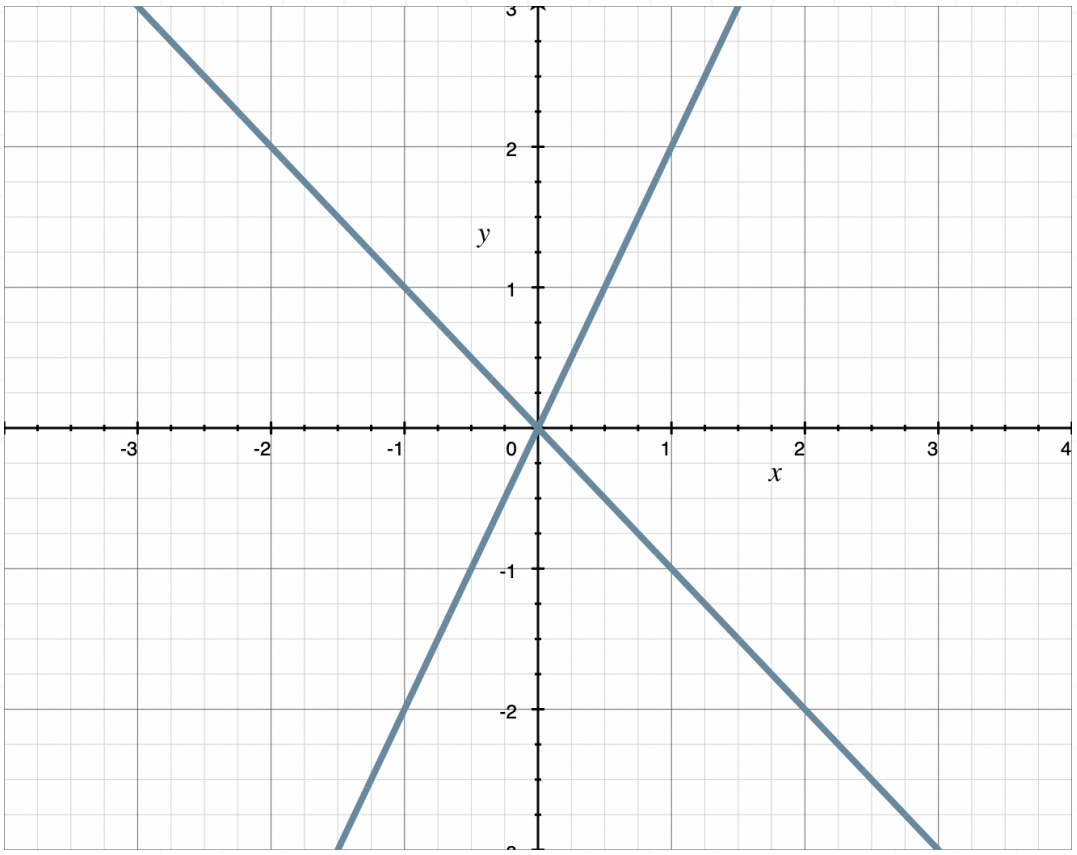
$$\lambda = -1, 5$$

gives the Eigenvalues $\lambda_1 = -1$ and $\lambda_2 = 5$, and their associated Eigenvectors

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \qquad \overrightarrow{k_2} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

The Eigenvector $\overrightarrow{k_1} = (1, -1)$ lies along the line $y = -x$, and the Eigenvector $\overrightarrow{k_2} = (1,2)$ lies along the line $y = 2x$, so we'll sketch these lines.

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The Eigenvalue associated with $\overrightarrow{k_1} = (1, -1)$ is $\lambda = -1$, which means the direction along that trajectory is toward the origin. The Eigenvalue associated with $\overrightarrow{k_2} = (1,2)$ is $\lambda = 5$, which means the direction along that trajectory is away from the origin.

Because the Eigenvalues have opposite signs, we're dealing with an unstable saddle point.

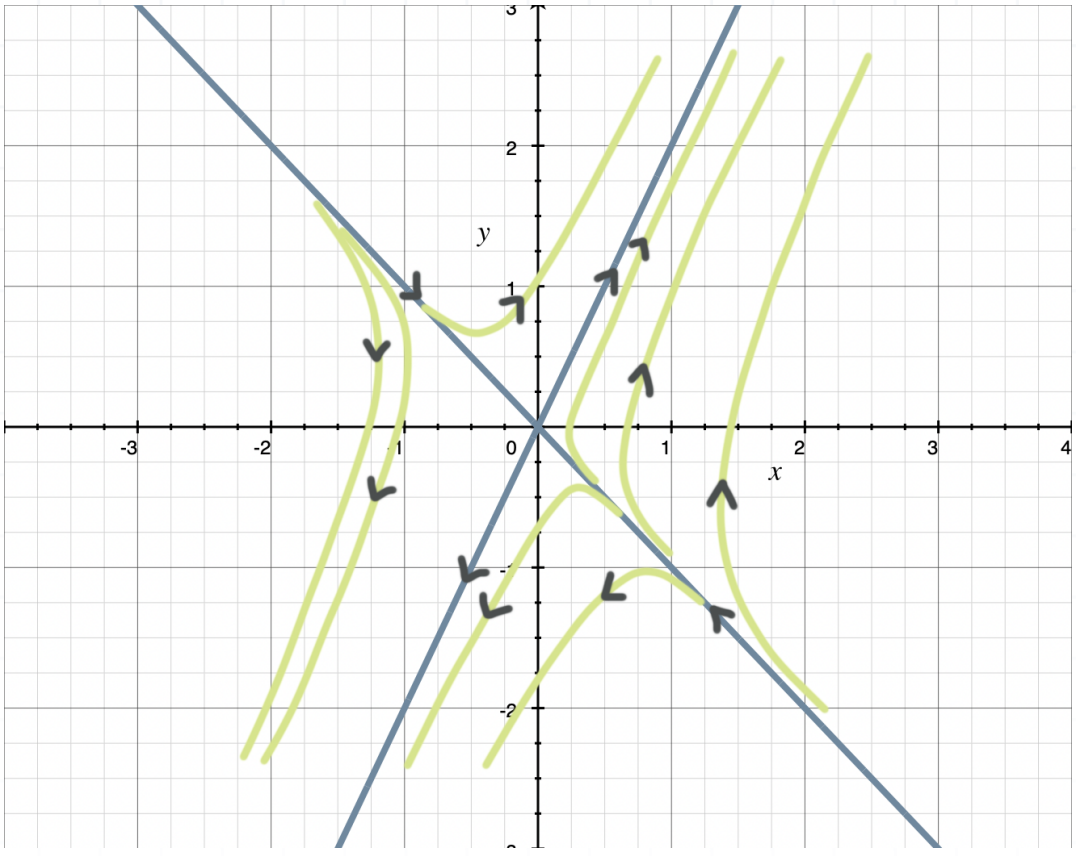
To apply the $t \to \pm\infty$ test, we'll use the general solution to the system,

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{5t}$$

Because e^{-t} dominates e^{5t} as $t \to -\infty$, the $\overrightarrow{k_2} = (1,2)$ vector drops away first, meaning that our trajectories are going to "start" parallel to $\overrightarrow{k_1} = (1, -1)$. On the other end, e^{5t} dominates e^{-t} as $t \to \infty$, so the $\overrightarrow{k_1} = (1, -1)$ vector will drop away first, meaning that our trajectories are going to "end" parallel to $\overrightarrow{k_2} = (1,2)$.

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Therefore, starting the trajectories parallel to $\overrightarrow{k_1} = (1, -1)$ and ending them parallel to $\overrightarrow{k_2} = (1,2)$ means that the phase portrait must look something like



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